

Travel Planner - PostgreSQL / pgAdmin4 Setup Guide

1. Install PostgreSQL + pgAdmin4

Download and install PostgreSQL:

- <https://www.postgresql.org/download/>

During installation:

- Remember your PostgreSQL password
- Keep the default port:
- 5432

pgAdmin4 is usually installed automatically with PostgreSQL.

2. Create the Database

Open:

pgAdmin4

Then:

1. Expand:

Servers
→ PostgreSQL
→ Databases

1. Right click:

Databases
→ Create
→ Database

1. Create database:

travel_planner

1. Press:

Save

3. Open Query Tool

Right click:

travel_planner
→ Query Tool

4. Run the Full Schema

Copy and run ALL of the SQL below.

```
CREATE TABLE users (  
    user_id SERIAL PRIMARY KEY,  
    username VARCHAR(255) UNIQUE NOT NULL,  
    password VARCHAR(255) NOT NULL,  
    role VARCHAR(50) DEFAULT 'USER',  
    uid VARCHAR(20) UNIQUE  
);  
  
CREATE TABLE trips (  
    trip_id SERIAL PRIMARY KEY,  
    trip_name VARCHAR(255) NOT NULL,  
    destination VARCHAR(255) NOT NULL,  
    start_date VARCHAR(50) NOT NULL,  
    end_date VARCHAR(50) NOT NULL,  
    owner_email VARCHAR(255)  
);  
  
CREATE TABLE itinerary_items (  
    item_id SERIAL PRIMARY KEY,  
    trip_id INTEGER REFERENCES trips(trip_id) ON DELETE CASCADE,  
    title VARCHAR(255) NOT NULL,  
    item_date VARCHAR(50) NOT NULL,  
    item_time VARCHAR(50) NOT NULL,  
    location VARCHAR(255)  
);  
  
CREATE TABLE friend_requests (  
    request_id SERIAL PRIMARY KEY,  
    sender_id INTEGER REFERENCES users(user_id) ON DELETE CASCADE,  
    receiver_id INTEGER REFERENCES users(user_id) ON DELETE CASCADE,
```

```

        status VARCHAR(20) DEFAULT 'PENDING'
    );

CREATE TABLE friends (
    friendship_id SERIAL PRIMARY KEY,
    user1_id INTEGER REFERENCES users(user_id) ON DELETE CASCADE,
    user2_id INTEGER REFERENCES users(user_id) ON DELETE CASCADE
);

CREATE TABLE shared_trips (
    shared_trip_id SERIAL PRIMARY KEY,
    trip_id INTEGER REFERENCES trips(trip_id) ON DELETE CASCADE,
    owner_id INTEGER REFERENCES users(user_id) ON DELETE CASCADE,
    shared_with_id INTEGER REFERENCES users(user_id) ON DELETE CASCADE,
    can_edit BOOLEAN DEFAULT FALSE
);

```

Press:

Execute Script

or:

F5

5. Generate User UIDs Automatically

Run this SQL AFTER the main schema:

```

CREATE OR REPLACE FUNCTION generate_uid()
RETURNS TRIGGER AS $$
BEGIN
    NEW.uid := 'USR' || LPAD(NEW.user_id::TEXT, 5, '0');
    RETURN NEW;
END;
$$ LANGUAGE plpgsql;

CREATE TRIGGER user_uid_trigger
AFTER INSERT ON users
FOR EACH ROW
EXECUTE FUNCTION generate_uid();

```

Press:

F5

6. Update DatabaseConnection.java

Open:

src/main/java/com/travelplanner/DatabaseConnection.java

Update:

```
private static final String URL =  
    "jdbc:postgresql://localhost:5432/travel_planner";  
  
private static final String USER =  
    "postgres";  
  
private static final String PASSWORD =  
    "YOUR_POSTGRES_PASSWORD";
```

Replace:

YOUR_POSTGRES_PASSWORD

with your own PostgreSQL password.

7. Run the Application

Open the project in IntelliJ.

Run:

Main.java

8. First Time Testing

Recommended test flow:

1. Register User 1

2. Register User 2
 3. Login as User 1
 4. Create a trip
 5. Add itinerary activities
 6. Send friend request
 7. Login as User 2
 8. Accept request
 9. Share trip
 10. Test VIEW ONLY permissions
 11. Test CAN EDIT permissions
-

9. Important Notes

DO NOT SHARE:

- PostgreSQL passwords
 - `.idea/`
 - `target/`
 - `out/`
-

10. Recommended .gitignore

```
# Build files
target/
out/
*.class
*.log

# IDE files
.idea/
.vscode/

# macOS
.DS_Store

# Legacy local storage files
users.txt
trips.txt

# Environment / secrets
.env
```

11. Common Errors

"Connection refused"

Make sure PostgreSQL server is running.

"password authentication failed"

Incorrect PostgreSQL password in:

```
DatabaseConnection.java
```

"database travel_planner does not exist"

You forgot to create the database.

"relation already exists"

The tables already exist.

This is safe.

12. Project Features

The project currently supports:

- User registration/login
 - PostgreSQL persistence
 - Trip creation
 - Itinerary management
 - Friend requests
 - Friends list
 - Shared trips
 - View-only permissions
 - Editable shared trips
 - Collaboration system
 - Modern JavaFX UI
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END OF README